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# Workshop 1: Multi-Armed Bandits

Exploration, Regret, and Algorithm Comparison

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ELEN90098 Reinforcement Learning for Engineers  
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Workshop oral assessment  
10 min presentation + 5 min Q&A

# Experimental setup

$$k = 10, \quad q_*(a) \sim \mathcal{N}(0, 1), \\ R_t \mid A_t = a \sim \mathcal{N}(q_*(a), 1).$$

**Fixed testbed, seed 5.** The optimal arm is  $a_* = 2$  in Python indexing (the third arm), with  $q_*(a_*) = 2.4308$ .

$$T = 10^4$$

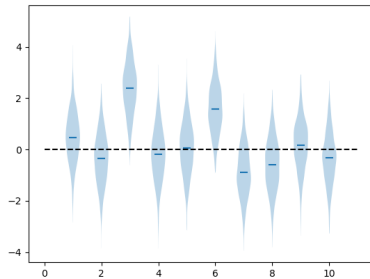
Q4–Q7 horizon

Q3 growth diagnostics extend to  $T = 10^5$  with  $M = 500$ .

The same fixed  $q_*$  is used throughout.

$$M = 300$$

independent reruns



The simulator knows  $q_*$  only to evaluate pseudo-regret; the agents do not observe it.

# What does sublinear regret mean?

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$$\text{Regret}_T = \mathbb{E} \left[ \sum_{t=0}^{T-1} \left( q_*(a_*) - q_*(A_t) \right) \right],$$

$$\text{Regret}_T = o(T) \iff \lim_{T \rightarrow \infty} \frac{\text{Regret}_T}{T} = 0.$$

**Average opportunity loss vanishes, even though cumulative regret may still grow.**

Simulation estimator:  $\widehat{R}_T = M^{-1} \sum_{r=1}^M \sum_{t=0}^{T-1} [q_*(a_*) - q_*(A_t^{(r)})]$ .

## Why it matters

- The policy spends a vanishing fraction of time on inferior arms.
- Exploration can continue, but its cumulative cost must grow slower than  $T$ .
- A straight-looking finite curve is not enough: the growth rate must be tested across horizons.

# How do we test sublinearity numerically?

## 1. Normalised regret

$$\hat{\rho}_T = \frac{\hat{R}_T}{T}$$

Sublinear behaviour requires  $\hat{\rho}_T$  to trend toward zero as the horizon grows.

## 2. Incremental slope

$$\hat{s}_i = \frac{\hat{R}_{T_{i+1}} - \hat{R}_{T_i}}{T_{i+1} - T_i}$$

A falling late-horizon slope indicates decreasing marginal regret.

## 3. Effective exponent

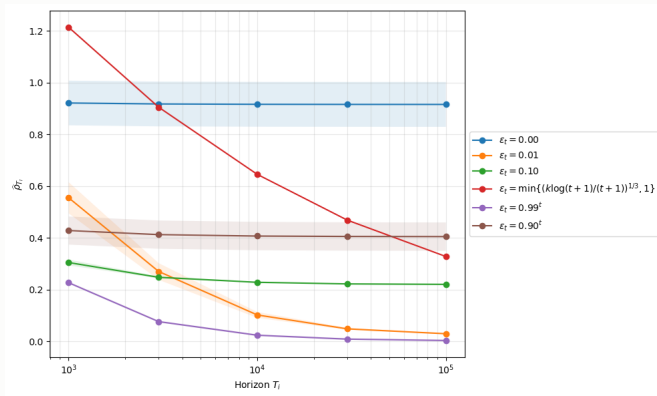
$$\hat{\alpha}_i = \frac{\log \hat{R}_{T_{i+1}} - \log \hat{R}_{T_i}}{\log T_{i+1} - \log T_i}$$

Values below one are consistent with  $\hat{R}_T \propto T^\alpha$  over that interval.

**Protocol.** Evaluate a common set of horizons  $\{10^3, 3 \times 10^3, 10^4, 3 \times 10^4, 10^5\}$ , average over 500 reruns, and examine all three diagnostics together.

These are finite-horizon diagnostics. They provide evidence consistent with or against sublinearity; they do not prove an asymptotic theorem.

# Which schedules show sublinear regret?



At  $T = 10^5$

| schedule          | $\hat{\rho}_T$ |
|-------------------|----------------|
| $\epsilon = 0$    | 0.9158         |
| $\epsilon = 0.01$ | 0.0300         |
| $\epsilon = 0.10$ | 0.2207         |
| polynomial        | <b>0.3283</b>  |
| $0.99^t$          | 0.0040         |
| $0.90^t$          | 0.4052         |

Polynomial decay continues to fall with late slope 0.2681.  
The very small mean for  $0.99^t$  needs a run-level audit.

# Why the exploration schedule matters

## Too little exploration

$\varepsilon = 0$  and rapidly decaying  $0.90^t$  can commit early to a wrong arm. Their late slopes remain 0.9157 and 0.4049.

## Persistent fixed exploration

Fixed  $\varepsilon$  keeps sampling inferior arms. The positive regret rate is small for 0.01 but does not vanish in this experiment.

## Decaying but persistent

$\varepsilon_t = \min\{(k \log(t+1)/(t+1))^{1/3}, 1\}$  balances continued discovery with decreasing exploration cost.

# 499/500

$0.99^t$  runs with essentially zero late slope

# 1/500

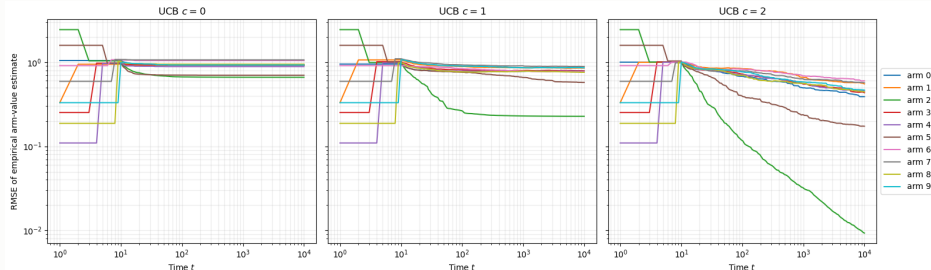
run with late slope 0.8483

## A low mean can hide a rare catastrophic run.

Because  $\sum_t 0.99^t < \infty$ , exploration eventually becomes negligible; an early wrong commitment can then persist.

Observed positive-run proportion: 0.002. This is descriptive evidence of a lock-in mechanism, not an estimate of its asymptotic probability.

# UCB arm-value estimation



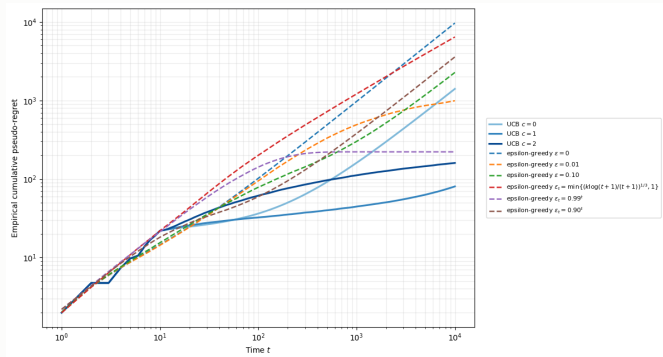
## Final mean arm-wise RMSE

| UCB $c$ | mean RMSE     |
|---------|---------------|
| 0       | 0.8981        |
| 1       | 0.7254        |
| 2       | <b>0.4090</b> |

Per arm,  $\text{RMSE}_T(a) = \sqrt{M^{-1} \sum_r (Q_T^{(r)}(a) - q_*(a))^2}$ ; the table averages this over ten arms.

**Result:** larger  $c$  estimates all arms better. This differs from minimising regret.

# UCB versus epsilon-greedy regret



**Regret at  $T = 10^4$**

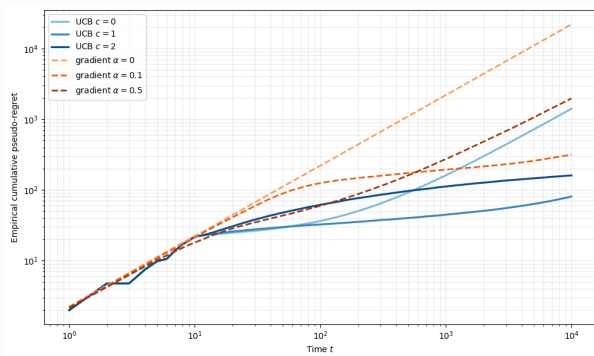
| policy                | $\hat{R}_T$  |
|-----------------------|--------------|
| UCB $c = 1$           | <b>80.58</b> |
| UCB $c = 2$           | 160.29       |
| $\epsilon_t = 0.99^t$ | 221.98       |
| $\epsilon = 0.01$     | 994.19       |

**Lowest regret: UCB**  
 **$c = 1$ . Lowest RMSE:**  
**UCB  $c = 2$ .**

Index:

$$Q_t(a) + c \sqrt{\log(t+1)/N_t(a)}.$$

# Gradient bandit versus UCB



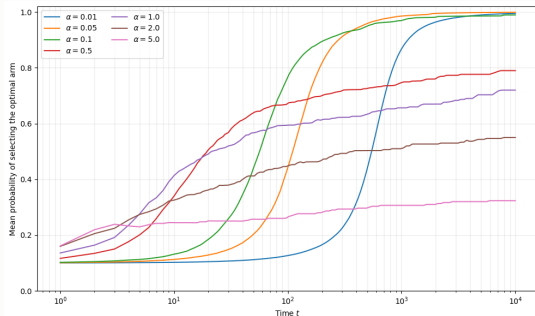
Regret at  $T = 10^4$

| policy               | $\hat{R}_T$  |
|----------------------|--------------|
| UCB $c = 1$          | <b>80.58</b> |
| UCB $c = 2$          | 160.29       |
| Grad. $\alpha = 0.1$ | 314.58       |
| Grad. $\alpha = 0.5$ | 1968.16      |
| UCB $c = 0$          | 1408.07      |

**3.90 $\times$  more regret**

Best gradient versus UCB  $c = 1$ ,  
on this experiment.

# Gradient step-size sensitivity



## Final performance at $T = 10^4$

| $\alpha$ | mean $P_T(a_*)$ | $\hat{R}_T$   |
|----------|-----------------|---------------|
| 0.01     | 0.9952          | 1542.45       |
| 0.05     | <b>0.9991</b>   | 357.39        |
| 0.10     | 0.9896          | <b>314.58</b> |
| 0.50     | 0.7899          | 1968.16       |
| 2.00     | 0.5500          | 5358.04       |
| 5.00     | 0.3233          | 11148.71      |

At  $\alpha = 5$ : 32.33% of runs end above 0.99, 67.67% below 0.01, and none in between.

Our results do not support optimal-arm convergence for arbitrary  $\alpha > 0$ .

# Conclusions and questions

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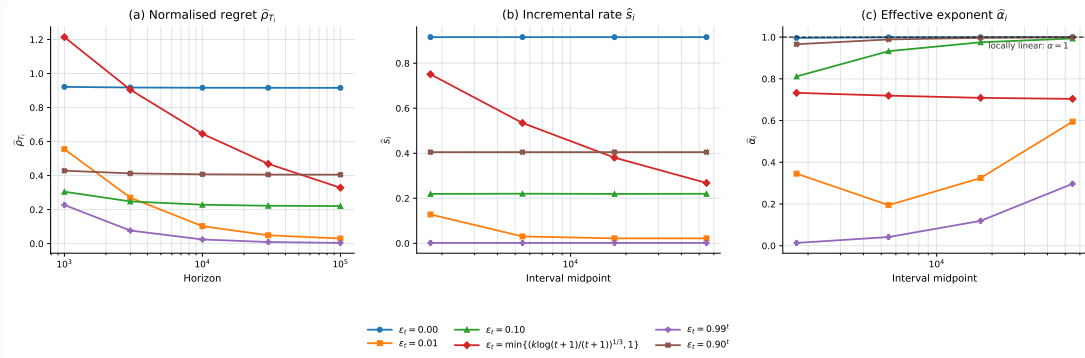
1. **Q1–Q2:** sublinear regret means vanishing regret per step; credible finite-horizon evidence needs several complementary diagnostics.
2. **Q3:** polynomially decaying exploration gives the clearest sublinear trend; a low average can conceal rare lock-in.
3. **Q4–Q5:** UCB  $c = 2$  estimates arm values best, while UCB  $c = 1$  achieves the lowest regret.
4. **Q6–Q7:** gradient bandit works well only in a moderate step-size range; aggressive updates can create bimodal outcomes.

All reported comparisons use the same fixed ten-arm Gaussian testbed and notebook protocols. Claims are empirical unless stated otherwise.

## Questions?

5-minute discussion

# Additional epsilon-greedy diagnostics



The three diagnostics support the same qualitative interpretation;  $\hat{\alpha}_i$  is supplementary, while  $\hat{\rho}_{T_i}$  and  $\hat{s}_i$  are the primary numerical tests.

# Rare-failure audit for $\varepsilon_t = 0.99^t$

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Run-level result, 30,000 to 100,000

**499**

near-zero-slope runs

**1**

run at slope 0.84829

| statistic             | value    |
|-----------------------|----------|
| mean late slope       | 0.001697 |
| positive-run fraction | 0.002    |
| positive-run slope    | 0.84829  |

Mechanism

$$\sum_{t=0}^{\infty} 0.99^t = 100 < \infty.$$

- Only a finite expected amount of random exploration is scheduled.
- Most runs identify the optimal arm before exploration disappears.
- A rare early error can survive and then accumulate near-linear regret.

# UCB arm-level estimates at $T = 10,000$

| arm | $q_*(a)$ | mean $Q_T(a)$ , $c = 2$ | mean $N_T(a)$ |
|-----|----------|-------------------------|---------------|
| 0   | 0.441    | 0.316                   | 9.3           |
| 1   | -0.331   | -0.459                  | 5.3           |
| 2   | 2.431    | 2.430                   | 9900.0        |
| 3   | -0.252   | -0.321                  | 5.7           |
| 4   | 0.110    | -0.004                  | 7.2           |
| 5   | 1.582    | 1.526                   | 44.0          |
| 6   | -0.909   | -1.058                  | 3.8           |
| 7   | -0.592   | -0.755                  | 4.4           |
| 8   | 0.188    | 0.090                   | 7.6           |
| 9   | -0.330   | -0.423                  | 5.3           |

## Mean arm-wise RMSE

| $c$ | RMSE          |
|-----|---------------|
| 0   | 0.8981        |
| 1   | 0.7254        |
| 2   | <b>0.4090</b> |

Even  $c = 2$  allocates almost all pulls to the best arm. Its extra exploration primarily improves estimates of the remaining arms.

# Full gradient step-size sweep

| $\alpha$ | mean $P_T(a_*)$ | regret   | $P_T > 0.99$ | $P_T < 0.01$ |
|----------|-----------------|----------|--------------|--------------|
| 0.01     | 0.9952          | 1542.45  | 1.0000       | 0.0000       |
| 0.05     | 0.9991          | 357.39   | 1.0000       | 0.0000       |
| 0.10     | 0.9896          | 314.58   | 0.9900       | 0.0100       |
| 0.50     | 0.7899          | 1968.16  | 0.7900       | 0.2100       |
| 1.00     | 0.7200          | 2765.23  | 0.7200       | 0.2800       |
| 2.00     | 0.5500          | 5358.04  | 0.5500       | 0.4500       |
| 5.00     | 0.3233          | 11148.71 | 0.3233       | 0.6767       |

## Why “between” is zero

The final-policy distribution is almost binary: each run ends either strongly on the optimal arm or strongly on another arm. Therefore

$$\Pr(0.01 \leq P_T(a_*) \leq 0.99) = 0$$

for every tested  $\alpha$ . This column measures concentration, not statistical uncertainty.

# Algorithms used in the comparison

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## Epsilon-greedy

With probability  $1 - \varepsilon_t$ , select a greedy action; otherwise sample uniformly.

Values use sample-average updates.

## Upper confidence bound

$$A_t = \arg \max_a \left[ Q_t(a) + c \sqrt{\frac{\log(t+1)}{N_t(a)}} \right].$$

The bonus favours actions with uncertain value estimates.

## Gradient bandit

$$\pi_t(a) = \frac{\exp(H_t(a) - \max_b H_t(b))}{\sum_b \exp(H_t(b) - \max_j H_t(j))}.$$

$$\Delta H_t(a) = \alpha(R_t - \bar{R}_t)(\mathbf{1}_{a=A_t} - \pi_t(a)).$$

Preferences are updated by reward advantage relative to the running-average baseline. The shifted softmax is numerically stable.